This answer was also pedagogically useful (and has provided grist for the mills of many a science fiction writer). However, like the Oppenheimer–Snyder prediction, it has nothing to do with the real Universe in which we live because it is unstable against small perturbations. More specifically, in our real Universe, the black hole is continually bombarded by tiny electromagnetic vacuum fluctuations and by tiny amounts of radiation. As these fluctuations and radiation fall into the hole, the hole's gravity accelerates them to enormous energy, and they then explosively hit and destroy the little closed universe, just before the little universe begins its trip. This was conjectured by Penrose in 1968, and has since been verified in many different calculations, carried out by many different physicists.

Belinsky, Khalatnikov, and Lifshitz have given us yet another answer to our questions, and this one, being totally stable against small perturbations, is probably the "right" answer, the answer that applies to the real black holes that inhabit our Universe: *The star that forms the hole and everything that falls into the hole when the hole is young get torn apart by the tidal gravity of a BKL singularity*. (This is the kind of singularity that Belinsky, Khalatnikov, and Lifshitz discovered, as a solution of Einstein's equation, after Penrose convinced them that singularities must inhabit black holes.)

The tidal gravity of a BKL singularity is radically different from that of the Oppenheimer–Snyder singularity. The Oppenheimer–Snyder singularity stretches and squeezes an infalling astronaut (or anything else) in a steady but mounting way; the stretch is always radial, the squeeze is always transverse, and the strengths of stretch and squeeze grow steadily and smoothly (Figure 13.1). The BKL singularity, by contrast, is somewhat like the taffy-pulling machines that one sometimes sees in candy stores or at carnivals. It stretches and squeezes first in this direction, then that, then another, then another, and yet another. The stretch and squeeze oscillate with time in a random and chaotic way (as measured by the infalling astronaut), but on average they get stronger and stronger, and their oscillations get faster and faster as the astronaut gets closer and closer to the singularity. Charles Misner (who discovered this type of chaotically oscillating singularity independently of Belinsky, Khalatnikov, and Lifshitz) has called this a *mixmaster oscillation* because one can imagine it mixing up the astronaut's body parts in the way that a mixmaster or eggbeater mixes up the yolk and white of an egg. Figure 13.6 depicts a specific